Fiscal Year:	FY 2015	**Task Last Updated:**	FY 06/22/2015
PI Name:	Kiss, John Z Ph.D.		
Project Title:	Novel Explorations Into the Interactions Between Light and Gravity Sensing in Plants		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
PI Email:	jzk@fit.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	336-334-5241
Organization Name:	University of North Carolina-Greensboro		
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City:	Greensboro	**State:**	NC
Zip Code:	27402-6170	**Congressional District:**	13
Comments:	NOTE: PI moved to Florida Institute of Technology from University of North Carolina-Greensboro in May 2024 (Ed., 2/6/25). NOTE: PI moved to University of North Carolina-Greensboro from University of Mississippi in July 2016 (Ed., 6/13/17) NOTE: PI moved to U Miss from Miami (Ohio) University in September 2012 (Ed., 8/10/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	09/11/2012	**End Date:**	12/31/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Taylor, Elizabeth	**Contact Phone:**	650.604.1783
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date is now 12/31/2016, per NSSC information (Ed., 6/15/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Edelmann, Richard (Miami University)		
Grant/Contract No.:	NNX12AO65G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	[ED. NOTE 9/28/2012: Continuation of "Novel Explorations Into the Interactions Between Light and Gravity Sensing in Plants"--grant # NNX10AM86G, per PI move from Miami-Ohio to University of Mississippi. See that project for previous reports] The major goals of this project are to determine how gravity and light responses influence each other in plants and to better understand the cellular signaling mechanisms involved in plant tropisms. This proposed project builds on our previous spaceflight experiments on the International Space Station (ISS) with the European Modular Cultivation System (EMCS) using the model plant Arabidopsis. In this new proposed project, we plan to confirm and extend our discovery during the recent ISS experiment of a novel red-light-based positive phototropic response in plant shoots. Our hypothesis is that positive red-light-sensing, which was known in older plant lineages, is masked by normal 1-g conditions in more recently evolved lineages. We also propose to investigate fundamental interactions among red and blue light signaling pathways and the gravity sensing mechanisms. The ISS is the only laboratory facility where it is possible to reliably and simultaneously vary light and gravity stimuli for the proposed experiments. These experiments can be accomplished with telemetric science and therefore fit with the extremely limited down mass capabilities of current spaceflight opportunities. The EMCS is an automated facility and can be operated largely with telemetry. Experimental containers will launched via Space-X, and during an experimental run, images of seedling growth responses can be downlinked to Earth. A significant advantage of this proposed project is that we will use existing flight hardware that has been flown successfully and allowed for growth of Arabidopsis seedlings in the EMCS. This project is also relevant to the recent emphasis at NASA to study plant growth and development at fractional g-levels such as those found on the moon and Mars. Improved knowledge of the basic mechanistic processes that will be the focus of this project is vital to develop ways to use plants in extraterrestrial bioregenerative life support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We plan to study the effects of gravity and light on basic cellular mechanisms in plants, so this project is relevant to the emphasis on using Arabidopsis thaliana as a model system in developmental biology. Since plants will be a necessary part of bioregenerative life support needed to send humans to Mars and beyond, the knowledge obtained from our spaceflight experiments will be critical for developing ways to effectively use plants in the bioregenerative life support systems. These life support systems also can be used in ground-based Earth applications such as the design of large-scale greenhouses to support crop plants and agriculture. In addition, improved knowledge of light and gravity systems in plants will, in the long term, allow us to develop better crop plants for use on Earth.
Task Progress:	1. September 21, 2014; Kennedy Space Center (KSC): Launch of Space-X4 with Seedling Growth-2 (SG-2). 2. October – November 2014; ISS: Completion of three experimental runs of SG-2. Images were downlinked. Seedlings were frozen in MELFI and stored on ISS until the Space X-5 mission. 3. October 2014; Pasadena: Presentation by PI J.Z. Kiss of preliminary results from Seedling Growth (SG) experiments at American Society for Gravitational and Space Research (ASGSR) annual meeting. 4. November 2014; Pisa, Italy: Presentation by PIs of preliminary results from SG at a European Space Symposium. 5. December 2014; ISS: Completion of addition fourth run of SG-2 (due to incomplete hydration in one of the ECs). 6. January 2015; ISS: transfer of SG-2 frozen samples from ISS MELFI to SpaceX-5 GLACIER. 7. February 2015; Return of Space-X5 with the frozen seedlings from Seedling Growth-1 (SG-1). 8. March 2015; Ames Research Center (ARC); SG-2 samples were retrieved from hardware and shipped to Javier Medina’s lab in Spain. 9. May 2015; International Working Group in Madrid to discuss progress in the SG series of space experiments.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Abstracts for Journals and Proceedings	Kiss JZ, Millar KDL, Edelmann RE, Herranz R, Medina FJ. "Phototropism experiments in microgravity--the Seedling Growth project in the EMCS on the ISS." Committee on Space Research (COSPAR) 2014 40th Scientific Assembly, Moscow, Russia, August 2-10, 2014. Abstract book. Committee on Space Research (COSPAR) 2014 40th Scientific Assembly, Moscow, Russia, August 2-10, 2014. , Aug-2014
Abstracts for Journals and Proceedings	Medina FJ, Valbuena MA, Youssef K, Kiss JZ, van Loon JJWA, Herranz R. "Meristematic competence is disrupted by microgravity, real or simulated, in seedlings and cultured cells of Arabidopsis." Committee on Space Research (COSPAR) 2014 40th Scientific Assembly, Moscow, Russia, August 2-10, 2014. Committee on Space Research (COSPAR) 2014 40th Scientific Assembly, Moscow, Russia, August 2-10, 2014. Abstract Book. , Aug-2014
Abstracts for Journals and Proceedings	Kiss JZ, Vandenbrink JP, Millar KDL, Edelmann RE, Medina FJ, Herranz R. "Examination of plant tropisms and the cell cycle in microgravity." 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Abstracts for Journals and Proceedings	Kiss JZ. "Plant tropisms: from Darwin to the International Space Station." Symposium--Biology in Space: Challenges and Opportunities, Pisa, Italy, November 2014. Symposium--Biology in Space: Challenges and Opportunities, Pisa, Italy, November 2014. , Nov-2014

Abstracts for Journals and Proceedings	Neel A, Vandenbrink JP, Kiss JZ. "The effects of red-light pretreatment on blue-light phototropism." American Society of Plant Biologists (Southern Section) meeting, Dauphin Island, AL, March 28-30, 2015. American Society of Plant Biologists (Southern Section) meeting, Dauphin Island, AL, March 28-30, 2015. , Mar-2015
Abstracts for Journals and Proceedings	Vandenbrink JP, Herranz R, Medine FJ, Edelmann RE, Kiss JZ. "The effects of reduced gravity on phototropism in response to red or blue light." American Society of Plant Biologists (Southern Section) meeting, Dauphin Island, AL, March 28-30, 2015. American Society of Plant Biologists (Southern Section) meeting, Dauphin Island, AL, March 28-30, 2015. , Mar-2015
Articles in Other Journals or Periodicals	Kiss JZ. "Book review of 'Why Mars: NASA and the Politics of Space Exploration.' By W.H. Lambright." Choice 2014 Dec;52:1962. , Dec-2014
Articles in Other Journals or Periodicals	Kiss JZ. "Book review of 'Bold They Rise: The Space Shuttle Early Years, 1972-1986.' By D. Hitt, H.R. Smith." Choice 2015 Feb;52:3081. , Feb-2015
Articles in Other Journals or Periodicals	Kiss JZ. "Book review of 'Sally Ride: America's First Woman in Space.' By L. Sherr." Choice 2015 Mar;52:3635. , Mar-2015
Articles in Other Journals or Periodicals	Kiss JZ. "Book review of 'Rockets and revolution: a cultural history of early spaceflight.' By M.G. Smith." Choice 2015 May;52:4736. , May-2015
Articles in Peer-reviewed Journals	Sindelar TJ, Millar KDL, Kiss JZ. "Red-light effects on blue-light-based phototropism in roots of Arabidopsis thaliana." International Journal of Plant Sciences. 2014 Jul-Aug;175(6):731–40. http://dx.doi.org/10.1086/676303 , Jul-2014
Articles in Peer-reviewed Journals	Vandenbrink JP, Kiss JZ, Herranz R, Medina FJ. "Light and gravity signals synergize in modulating plant development." Frontiers in Plant Science. 2014 Oct 28;5:563. eCollection 2014. http://dx.doi.org/10.3389/fpls.2014.00563 ; PubMed PMID: 25389428; PubMed Central PMCID: PMC4211383 , Oct-2014
Articles in Peer-reviewed Journals	Kiss JZ. "Conducting plant experiments in space." Methods Mol Biol. 2015;1309:255-83. http://dx.doi.org/10.1007/978-1-4939-2697-8_19 ; PubMed PMID: 25981781 , May-2015
Awards	Kiss JZ. "NASA Outstanding Public Leadership Medal 'for exceptional contributions in spaceflight research in the fundamental biology of plants in support of NASA's exploration mission.' November 2015." Nov-2015
Significant Media Coverage	Kiss JZ. Interview. "Space Station Live: Seedling Growth-2 Experiment." NASA TV on YouTube: September 24, 2014. https://youtu.be/lnVque_-f7U (accessed 6/25/15). NASA Public Affairs Officer Lori Meggs interviews John Kiss, the principal investigator for the Seedling Growth-2., Sep-2014